

# MOHD ANAS

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## EDUCATION

### B.Tech, Information Technology

IIIT Vadodara, Gujarat, India

Expected Graduation: Jan 2027

7.43 / 10.0 GPA

**Coursework:** Backend System, Distributed System, Operating Systems, Object-Oriented Programming (OOP), Design Patterns, Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Automation

## PROFESSIONAL EXPERIENCE

### Aarogya Kaya | SDE Intern | Remote

July 2025 - Jan 2026

- Architected Go Backend services for Shiprocket and Instagram Graph API integrations, implementing **CQRS architecture** and a **three-tier caching strategy (in-memory, Redis, and database)**, reducing average read latency from 420ms to 38ms (91%) and maintaining p95 response times below 250ms.
- Refactored 15+ Backend modules using **SOLID principles** and **Design Patterns** (such as **Dependency Injection**), reducing code duplication by 30% and improving feature delivery speed.
- Investigated and resolved **production issues** reported by customers, reducing turnaround time through structured debugging and monitoring.

### Open Source Contributor | Remote

July 2025 - Present

- Contributed a merged Pull Request to **Next.js (Vercel)**, migrating the patch-next utility from CommonJS to TypeScript ESM and reducing contributor setup overhead by eliminating repeated dependency reinstalls.
- Contributed **22 merged Pull Requests** and opened **12 Issues** across open-source projects, fixing bugs, implementing features, and improving developer tooling.

## PROJECTS

### a3.store | Distributed Key-value Store | Python, EKS, AWS, SQLite, Prometheus

Live | GitHub

- Developed a **leaderless** Distributed key-value store implementing replication, **failure detection**, synchronization, gossip-based membership, and anti-entropy repair across a 3-node cluster.
- Deployed the system on **AWS EKS** using Kubernetes StatefulSets and persistent volumes, managing rolling upgrades, pod restarts, and **self-healing behavior**.
- Implemented **operational dashboards** and **disaster-recovery workflows** to make the system debuggable, reliable, and usable in production-like environments.

### Second Brain | Career Comparison Simulation | Python, Redis Queue, Zustand, Render

Live | GitHub

- Reduced API latency by **65% (15s → 5.2s)** by offloading a 3-pass LLM pipeline to **Redis/RQ background workers** and optimizing **FastAPI + asyncpg** request handling.
- Built an interactive decision-graph visualizer supporting **100+ nodes** at **60 FPS** using **React Flow** and **Zustand**, enabling smooth zoom, pan, and branch expansion.
- Diagnosed structured-generation failure and introduced **Zod-enforced validation**, increasing valid JSON outputs to **98.2%**.

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, Go, C/C++, JavaScript, TypeScript, Rust, SQL, Node.js, React, Next.js, gRPC

**Tools & Technologies:** Git, GitHub, Docker, **Kubernetes**, **AWS**, EC2, EKS, IAM, VPC, Terraform, **Linux/Unix**, Kafka, Jenkins, Redis, SQLite, Prometheus, Grafana, Loki

**AI & LLMs:** HuggingFace, LangChain, PyTorch, NLP, Deep Learning, Generative AI

## ACHIEVEMENT

- Achieved Codeforces **Pupil (1245)**.
- **Solved 500+** Data Structures & Algorithms problems across Codeforces, LeetCode, CodeChef, and GeeksforGeeks.

## LEADERSHIP & ACTIVITIES

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Organized **5+** events and workshops, driving student engagement in **competitive programming** as **Coding Club Member**.
- Led recruitment, mentorship, and event initiatives for **100+ students** as **Joint Secretary of DOT Club**.